Homework 2: Research with Inheritance and Templates

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The videos explains more.

Overview

The motivation:

```
                Bag
              /     \
          List       ... (Stack, Queue ... )
          / |  \
         /  |   \
 CommonList  ...   ArrayList
```

Challenges

- So many pitfalls when using templates and inheritance.

Tasks

bag.h

list.h

A eager base class vs. a lazy base class
Base class describes the computation process.
step1
step2
step3...

while some steps can be decided by the derived classes.
how to call a function without defining the function?
algorithm(){
step1();
step2();
}

way 1: pure virtual function
Problem: constructor and destructor needs quite some computation.

way 2: use function pointer.
a) pointer to static funtion : just like a funtion
b) pointer to member function pointer: more complex ClassName::...

```
    a static function, like c function, needs all information/resource  in the parameter

    resources:  two parts:
    1) Data part  :  put all data resource in a struct.
    2) function/method part
       When a function pointer is a member of a struct/class, it cannot directly call ot
       put all method resource in struct.
```

common_list.h

test_common_list.cpp

Learned lessons and principles

- Virtual functions should not be called in constructors or destructors
  - Undefined behavior
- After a template class is declared/defined. When use this class, its name is
  The full name of a template SomeClass is:
  `template<typename T> someClass<T>`
  not
  `template<typename T> someClass`
- typedef does not work for template related types when some information is not totally decided.
- A declaration of static member inside a class is not a definition of the member. It needs a definition outside the class at some global scope.

Summary

Future research and exercise

- ArrayList
- Other container classes :

```
StackABC  <-----> ArrayStack
          |-----> ListStack

QueueABC <-----> ListQueue
         |----->CircularArrayQueue

ArrayABC <----> CommonArray
         |----> CircularArray

A common base class of Stack and Queue?
  StackQueueABS <-----> Stack
                |-----> Queue
```

References

[1] https://en.cppreference.com/w/cpp/language/override

[2] Why you must implement a template class in the header file?
https://wyfunique.github.io/posts/2019/02/blog-1-Implement_template_class_in_Cpp/
// good article.

[3] C++ template declarations vs. definitions
http://warp.povusers.org/programming/template_declarations.html
// shows how to declare a template class without definition.

[4] Can Virtual Functions be Inlined in C++?
https://www.geeksforgeeks.org/inline-virtual-function/
// virtual functions cannot be inlined.

[5] When to use const class member variables?
https://cplusplus.com/forum/beginner/285775/

[6] Why a virtual call to a pure virtual function from a constructor is UB and a call to a non-pure virtual function is allowed by the Standard?
https://stackoverflow.com/questions/9185990/why-a-virtual-call-to-a-pure-virtual-function-from-a-constructor-is-ub-and-a-cal
/* It mentions the notion of "virtual call". Also says that " a virtual call can never call pure virtual function". It means that pure virtual function can only be called oly with some fully-built object. Constructor and destructors are dealing with some intermediate object, not some fully built object */

[7] Pure virtual invocation from constructor and destructor
https://stackoverflow.com/questions/8651895/pure-virtual-invocation-from-constructor-and-destructor
/* Calling a pure virtual function from a constructor of destructor is forbidden or with undefined behavior. A reason could be that either the virtual table inside an object is not built yet, like when constructing an object, or already destroied, when deleting the object of a derived class. */

[8] "Direct" vs "virtual" call to a virtual function
https://stackoverflow.com/questions/36584670/direct-vs-virtual-call-to-a-virtual-function

[9] C++: call pure virtual function from member function of same class
https://stackoverflow.com/questions/30415669/c-call-pure-virtual-function-from-member-function-of-same-class

[10] Static member variables
https://www.learncpp.com/cpp-tutorial/static-member-variables/#:~:text=If no initializer is provided%2C static member variables are zero-initialized by default.
/* If no initializer is provided, static member variables are zero-initialized by default.

- Static members need to be defined outside the class because otherwise the compiler doesn't know which translation unit the static member should actually be defined in.
- With inline static members, we're telling the compiler to not worry about it and the linker to just pick one.
  */

[11] Static data members and templates (C++ only)
https://www.ibm.com/docs/en/zos/2.3.0?topic=only-static-data-members-templates-c

[12] static template functions in a class
https://stackoverflow.com/questions/9346076/static-template-functions-in-a-class

[13] Derived class friend of base class
https://bytes.com/topic/c/535409-derived-class-friend-base-class
/* You don't have to declare a derived class as a friend of it's base class. It is automatically a friend of it's base class. */

[14] Default constructor for a class with a reference data member?
https://stackoverflow.com/questions/22875186/default-constructor-for-a-class-with-a-reference-data-member
/* A class with a reference member needs to set the reference in its constructors. In most cases this means, that the class cannot have a default constructor. The best way to solve the problem is use a pointer instead of a reference: */

[15] Why static global variables initialized to zero, but static member variable in class not initialized?
https://stackoverflow.com/questions/74587411/why-static-global-variables-initialized-to-zero-but-static-member-variable-in-c
/* declaration inside the class is not a definition. Have to define a static int variable ouside the class. */

[16] Call static method in global scope
https://stackoverflow.com/questions/25660008/call-static-method-in-global-scope
/*
There are no "freestanding" function calls in C++ : you can call RegisterType() from another static method (e.g., from main())****
You can't call functions in the global scope of a file. It must have a scope that relates to the one of the main() function of your program
*/

[17] error "missing 'typename' prior to depen
https://cplusplus.com/forum/general/224845/

[18] Type alias, alias template (since C++11)
https://en.cppreference.com/w/cpp/language/type_alias
/* alias here only means for types, not an alias for variables. Declare reference as alias for variables. */

[19] Lambda expression in C++
https://www.geeksforgeeks.org/lambda-expression-in-c/

[20] static_cast和dynamic_cast详解
https://blog.csdn.net/u014624623/article/details/79837849
/* static cast does more checking than the C style type conversion. For casting from derived class to base class, static_cast and dynamic_cast are the same */

[21] Dynamic_cast and static_cast in C++

/* For dynamic_cast, You only need to use it when you're casting base class to a derived class. */

[22] Is it safe to delete[] of NULL?
cplusplus.com/forum/beginner/135030/
/* It is entirely safe to use delete or delete [] on a null pointer. */

[23] C++中NULL和nullptr的区别
/* https://blog.csdn.net/qq_18108083/article/details/84346655 /
/ NULL can be ambiguous, can be used as an integer 0, instead of a pointer. Therefore, nullptr will only be treated as a pointer, not an integer */

[24] Compiler Error C2683
https://learn.microsoft.com/en-us/cpp/error-messages/compiler-errors-2/compiler-error-c2683?view=msvc-170
You cannot use dynamic_cast to convert from a non-polymorphic class (a class with no virtual functions).

[25] Calling Member function in constructor
bytes.com/topic/c/579616-calling-member-function-constructor
/ * In a constructor, you should only call non-virtual member functions and static functions
of the class */

[26] Compiling error on template method, return is instance from inner class
https://stackoverflow.com/questions/8246117/compiling-error-on-template-method-return-is-instance-from-inner-class
/* When something is a dependent name, . i.e, some standalone type (without <...>) appears in a template, it could be an dependent name, unlike the types of int, double. The compiler needs to know that it is a type name, so you need to add the typename keyword before it. */

[27] Template Instantiation
https://docs.oracle.com/cd/E19205-01/819-5267/bkafi/index.html
/* good doc for all the c++ knowledge */

[28] How to implement a member function outside of a template class declaration body?
https://stackoverflow.com/questions/54440908/how-to-implement-a-member-function-outside-of-a-template-class-declaration-body
/* The full name of a template SomeClass is:
template someClass
not
template someClass
*/